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Got milk? The effect of export price shocks on exchange rates

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ABSTRACT

I examine the effect of exogenous terms of trade shocks on an exchange rate by turning to New Zealand's dairy auctions. Dairy is New Zealand's largest export category, making up almost 20 percent of exports. Specifically, whole milk powder accounts for 6 to 11 percent of total exports, and its price is determined in twice-monthly auctions. I use event studies to quantify the impact of surprise auction results on the New Zealand dollar on a high-frequency basis. I find that a 1 percent surprise increase in whole milk powder prices has a modest, but nevertheless significant, effect on the nominal exchange rate. The methodology developed here can potentially be applied to other commodity exporters.

1. Introduction

This paper provides novel evidence on the effect of terms of trade shocks on exchange rates using a high-frequency identification strategy amid a natural experiment. Theoretically, terms of trade increases should cause nominal exchange rate appreciations via the consumer price index (CPI) and price stickiness. However, while standard theory suggests a causal response of nominal exchange rates to terms of trade shocks, the actual effect has remained difficult to both identify and quantify empirically.

This effect is of interest especially to economies that rely on the export of a single commodity. How does the Chilean peso respond to copper price movements? How does the Russian ruble respond to oil price movements? Despite the widespread relevance of this question, many economies may struggle to identify a causal link if they focus internally. Because these countries often export a significant portion of world supply, they are not necessarily price takers. Thus, there may be confounding variables that affect both the price of the commodity via domestic supply and the exchange rate via the domestic CPI. In order to identify a causal link between an export commodity and an exchange rate, we need to be able to find and measure shocks to external demand for the commodity, which are exogenous to the domestic economy.

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I identify a setting that allows me to test the causal impact of such an export price shock on an exchange rate: New Zealand dairy auctions, specifically the auction of whole milk powder (WMP). Similar to other settings with a primary commodity export, these auctions involve a single product that comprises a large portion of the export price index. Dairy is New Zealand's primary export category, making up almost 20 percent of the country's exports, and WMP makes up 6 to 11 percent of total exports.² When dairy prices rise in New Zealand dollar terms, this should feed through to the export price index and the exchange rate with a magnitude large enough to detect. That said, similar to other countries for which a single commodity makes up a large percentage of the export basket (for example, copper in Chile or soy in Brazil), New Zealand *cannot* be considered a price taker in the world market for WMP; it accounts for over a third of world WMP production.³ Thus, we need an identification strategy in order to make any causal inference.

Two unique factors in the New Zealand setting allow me to measure exogenous shocks to its primary commodity. First, WMP prices are announced at precise times, and I can calculate surprise shocks in WMP prices at these times. Specifically, prices of various dairy products, including WMP, are determined at twice-monthly auctions. According to the efficient market hypothesis, changes in WMP prices should not impact the New Zealand dollar if these changes are expected and thus already priced into the New Zealand dollar. I thus focus on the extent to which auction prices differ from expectations. Since WMP futures cash-settle to the auction-determined price, the expected auction price can be calculated from the futures price right before the auction with very few assumptions. The difference between actual and expected prices can be substantial, with the actual price ranging from 19 percent below to 33 percent above the expected price.

Second, surprises at these auctions are due to factors exogenous to the New Zealand economy. Supply is announced a few days in advance, so surprises are determined by foreign demand. These demand shocks arise at the auction because a subset of buyers is excluded from the futures market due to capital controls and size limitations. Furthermore, I show that these demand shocks are dairy-specific and not indicative of changes to foreign income. These two ingredients make the New Zealand case unique. Results from this natural experiment may then inform our understanding of currencies from commodity-exporting countries in general.

Having identified an exogenous shock of sufficient magnitude, I calculate the response of the New Zealand dollar to this shock in a high-frequency setting. This allows me to isolate the effect of the price shock on the exchange rate, since noise is limited in such a narrow window. Within the 15 min after WMP price shocks, the exchange rate appreciates about 0.01 percent for a 1 percent surprise increase in WMP prices, which is directionally in line with theoretical expectations. This appreciation is statistically significant, and my study is the first to show a well-identified causal impact of terms of trade shocks on the exchange rate. While the magnitude is smaller than theory would suggest for a permanent shock, it can be reconciled under expectations of mean-reversion.

This paper fits into a few strands of the literature. First, it builds on the existing commodity currency literature (Gruen and Wilkinson, 1994; Amano and van Norden, 1995; Chen and Rogoff, 2003; Cashin et al., 2004; Clements and Fry, 2008; Tokarick, 2008; Chen et al., 2010; Powers, 2015). The empirical part of this literature uses time series methods to relate the terms of trade of a commodity-exporting country to its real exchange rate. For example, Chen and Rogoff (2003) look at how Australian, Canadian, and New Zealand real exchange rates vary with real non-energy commodity prices over a 25-year period, adjusting for various assumptions about the data-generating process. Similarly, Cashin et al. (2004) find evidence of a long-run relationship between the real exchange rate and real commodity prices for certain countries. In contrast, my paper seeks a sharper identification approach by examining the exchange rate around exogenous shocks to the export price index. Additionally, the existing literature finds conflicting results around whether commodity price changes induce exchange rate movements. For example, although Chen and Rogoff (2003) find that the Australia and New Zealand exchange rates move with the prices of their commodity exports, Chen et al. (2010) find insufficient evidence that country-specific commodity price indices Granger-cause movements in exchange rates. Notably, this literature tends to rely on a price-taker assumption of commodity exporters, which is often violated in practice. Because my paper relies on an identification strategy that is stronger than those previously used, my findings show that export price shocks do indeed cause exchange rate movements.

Second, my paper fits into the much larger exchange rate forecasting literature (Meese and Rogoff, 1983; Cheung et al., 2005; Engel et al., 2007; Gourinchas and Rey, 2007; Rossi, 2013). In general, this literature finds conflicting evidence on whether any of the standard theoretical exchange rate models translate into empirical exchange rate predictions. Recent papers (Lilley et al., 2022; Engel and Wu, 2023; Stavrakeva and Tang, 2024a; Chahrouh et al., 2024) have found that exchange rates, especially in the recent period, can be predicted with global risk appetite, forward guidance, US productivity, and the level of the exchange rate. Stavrakeva and Tang (2024b) find that macroeconomic news, including news on current account and trade balance, has large explanatory power for future exchange rates. In contrast, my study zooms in on terms of trade and seeks a sharp identification by using shorter windows.

Finally, my methods stem from the high-frequency identification literature. A large literature examines the high-frequency impact of monetary policy shocks (Kuttner, 2001; Gürkaynak et al., 2005; Hamilton, 2008; Campbell et al., 2012; Nakamura and Steinsson, 2018), while a smaller literature focuses on the effect of shocks on exchange rates (Andersen et al., 2003; Zettelmeyer, 2004; Faust et al., 2007). Unlike my paper, these papers do not look into terms of trade shocks; instead, they focus on either announcements of measures such as GDP and CPI or monetary policy announcements.

The rest of the paper is organized as follows. Section 2 provides a brief exchange rate model to fix ideas. Section 3 describes the dairy auctions and how they are relevant for the New Zealand economy. Section 4 describes my empirical approach in measuring the elasticity of the exchange rate to the terms of trade and provides evidence for critical identification assumptions. Section 5 presents the results of this elasticity identification. Section 6 concludes.

² More specifically, it made up 6 to 11 percent of total exports during the 2010–2019 period.

³ According to the Food and Agricultural Organization (FAO) of the United Nations, New Zealand produced between 26.2 and 37.5 percent of WMP globally between 2010 and 2019.

2. A brief exchange rate model

I start with a brief model of exchange rates in order to fix ideas on the drivers of exchange rate movements. This model comprises just two log-linearized equations, a country budget constraint and an Euler equation:⁴

$$\begin{aligned}\beta b_{t+1} - b_t &= nx_t = y_t^* - c_t \\ \mathbb{E}_t \Delta c_{t+1} &= 0\end{aligned}$$

where b_t is the country's net foreign asset position scaled by steady state exports, y_t^* denotes commodity exports, and c_t denotes imports (i.e., consumption of the tradable good). Per a Balassa-Samuelson framework, let the exchange rate be the ratio of non-tradable and tradable consumption:

$$e_t = y_t - c_t$$

where y_t is non-tradable consumption, which is equivalent to the country's production. Note that an increase in the exchange rate corresponds to a depreciation of the local currency.

For simplicity, assume that commodity exports y_t^* follow AR(1) processes with persistence parameter ρ_y . The budget constraint and Euler equation can be rolled forward and combined to solve for imports and, in turn, the exchange rate:

$$e_t = y_t - y_t^* \left(\frac{1 - \beta}{1 - \beta \rho_y} \right) - (1 - \beta) b_t \quad (1)$$

It is clear from this equation that the exchange rate appreciates when the terms of trade y^* increases. The elasticity of the exchange rate to the terms of trade depends on the discount rate and the persistence of the export process; if exports follow a random walk, this elasticity is unity.

In the following sections I describe a unique setting to test whether exchange rates do in fact move with terms of trade shocks. I will return to Eq. (1) when describing the results.

3. Setting and the relevance of GDT auctions

To understand how terms of trade shocks affect an exchange rate, I zoom in on a particular setting: the New Zealand dollar and Global Dairy Trade (GDT) auctions. GDT auctions were started in 2008 by Fonterra, New Zealand's largest dairy producer, in order to give buyers and sellers more transparency in pricing and create recognized benchmark prices for their products. The auctions are held twice a month and involve a variety of dairy items, including WMP. As discussed, dairy is New Zealand's largest export good category,⁵ with WMP alone accounting for 6 to 11 percent of goods and services exports over the course of my sample. This makes WMP New Zealand's single largest export, outside of the services sector, which is why I focus on WMP throughout this paper.⁶

The auctions do indeed serve their intended purpose of providing reference prices. As Fig. 1 shows, the price of WMP sold at GDT auctions is highly predictive of the WMP export price index published by New Zealand's official data agency. In fact, from 2010:Q4 through 2019:Q4 the two series were 88 percent correlated, adjusting for shipping lags.⁷ Furthermore, the price that Fonterra pays to its farmers, the farmgate milk price, is calculated using primarily GDT prices for the WMP, SMP, and AMF price inputs, again providing evidence for the relevance of GDT prices.⁸ As a final point, WMP auction prices exhibit a high degree of auto-correlation between auctions, with an autocorrelation coefficient of 0.968, indicating that auction prices reveal information about the future path of prices. For all of these reasons, market participants, including foreign exchange market participants, watch GDT auctions as an indicator of export prices, making it a relevant setting for studying shocks to New Zealand's terms of trade.⁹

⁴ I thank an anonymous referee.

⁵ From 2010 to 2014, dairy was the largest export category, including services. From 2015 to 2019, it was surpassed only by travel and tourism.

⁶ Specifically, I focus on the contract specification CP2 for regular WMP sourced in New Zealand. I also include skim milk powder (SMP) in robustness checks in Appendix B.

⁷ The lag appears because shipments occur after the auction date, and much of the visual difference in the two series is due to the fact that the WMP export price index is released quarterly, while the auction price is updated twice a month. Other discrepancies between the two series arise because not all WMP powder was sold at GDT auctions during this period. However, much of the off-auction sales happen in the 24 hours after the auction at the GDT auction price. In fact, all of Fonterra's non-GDT sales are linked to the GDT price, with different customers priced at a specified additive spread to the GDT price. This is economically significant in its own right, as Fonterra produces more than 80 percent of New Zealand's dairy, and anecdotally the same is true for certain other New Zealand WMP exporters.

⁸ From 2012 through 2016, GDT was the only price source for WMP, SMP, and AMF in the farmgate milk price calculation. Beginning in 2017, certain off-GDT sales were included; however, Fonterra notes in its 2019 farmgate milk price statement that including off-GDT sales increased the farmgate milk price by only 1.6 percent relative to not including these off-GDT sales.

⁹ For example, see *Wall Street Journal*, "New Zealand Dollar Down Late Eyes on Dairy Auction", June 15, 2015.

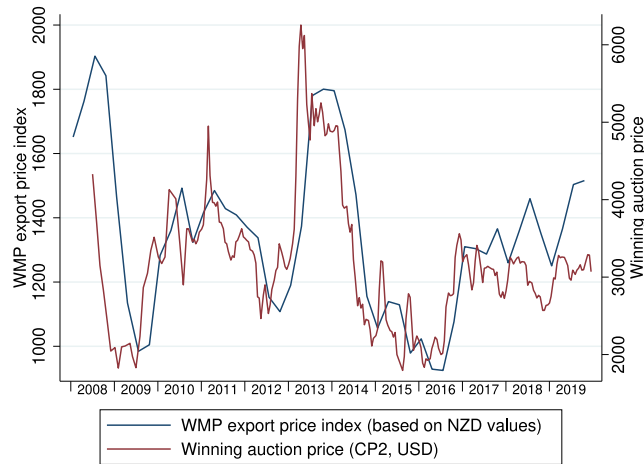


Fig. 1. Whole milk powder: auction price vs. export price index. This figure shows how the price of whole milk powder (WMP) sold in the Global Dairy Trade (GDT) auctions is highly predictive of the WMP export price index published by Statistics New Zealand in its Overseas Trade Indexes series. The auction price shown is the price of the largest contract specification sold in the GDT auctions (CP2, regular, produced in New Zealand) and is sourced from the Global Dairy Trade Historic Market Pack.

4. Empirical approach for elasticity identification

To estimate the extent to which WMP price movements affect the nominal exchange rate, I consider a regression of the form

$$\Delta e_a = \alpha + \beta \Delta p_a^{WMP} + \varepsilon_a, \quad (2)$$

where e is the log nominal exchange rate denominated in local currency per one unit of foreign currency, and p^{WMP} is the log price of whole milk powder in US dollars.¹⁰ Since export price increases should cause local currency appreciations (and since a positive movement in e indicates a local currency *depreciation*), we should expect β to be negative. To precisely identify the coefficient, I calculate Δe_a using minute-by-minute data in a tight window around GDT announcements. Results of the GDT auctions are not made public until the end of the auction, when prices are published on the GDT website.¹¹ Thus, prices are revealed at a specific point in time, and exchange rates should react accordingly at that time. While I would ideally know the exact time of result publication, these times are not available from GDT. I thus identify the time of price announcements by recording the first Bloomberg headline containing the auction results.¹² These Bloomberg headlines alert foreign exchange market participants as soon as GDT publishes the prices. This set-up embeds a joint hypothesis of both the importance of the WMP price shock and the importance of information dissemination by news outlets.¹³ Minute-by-minute exchange rate data come from histdata.com.¹⁴

My event window includes the five minutes before the announcement time through the 15 min after the announcement, in keeping with the literature (for example, Faust et al., 2007). Using such a narrow window reduces outside noise, allowing me to isolate the impact of price announcements on the exchange rate. I assume that all other variables are fixed during this window. I calculate the cumulative abnormal exchange rate return by comparing the cumulative returns during the event window with the average returns during my estimation window, the 60 min to 5 min before the announcement. I then compare this exchange rate response to the terms of trade shock. I specifically examine the NZD/AUD and NZD/USD exchange rates, since the United States and Australia are the two countries with floating currencies that import the most dairy from New Zealand.¹⁵ Summary statistics for the relevant exchange rates during the sample period are given in Table 1.

¹⁰ Although p^{WMP} should ideally be in local currency, WMP is sold in US dollars, so I will consider percentage changes in US dollar terms. However, when I convert $\hat{\beta}$ to reflect price changes in New Zealand dollar terms, the coefficient will be quite similar.

¹¹ Starting with the February 16, 2016, auction, GDT sold real-time access to auctions via its “Insights” package. I exclude auctions after this date in a robustness check in Appendix B.

¹² There are five days when the headline does not appear to be available on Bloomberg: February 18, 2014; March 4, 2014; April 1, 2014; January 2, 2019; and March 5, 2019. I exclude these days from my high-frequency analysis. Note also that starting with the September 16, 2014, auction, Bloomberg automated this headline release.

¹³ See Appendix D.

¹⁴ Note that histdata.com is a free, online datasources. I use it due to the availability of NZD/USD and NZD/AUD data at the one-minute frequency. However, the appendix discusses a benchmarking of the data using NZD/USD data from Refinitiv at the five-minute frequency. Additionally, as Table A.12 shows, using Refinitiv data does not change my baseline estimates.

¹⁵ Because WMP is such a small portion of these foreign economies, it does not necessarily matter which NZD exchange rate we look at, as we would not expect bilateral foreign exchange rates to move with a WMP price shock. For example, even if the CNY exchange rate were not fixed, we would not expect the CNY/USD exchange rate to move due to WMP auction announcements, so the CNY/NZD and USD/NZD exchange rate movements should be equivalent.

Table 1
Summary statistics for the New Zealand dollar.

	NZD/USD	NZD/AUD
Mean	1.352	1.138
Std. dev	0.124	0.097
Mean daily std. dev	0.0037	0.0020
Min	1.134	1.008
Max	1.612	1.367

Summary statistics are reported for the sample period October 8, 2010, through December 31, 2019, and are calculated using minute-by-minute data on trading days during the sample period. Mean daily standard deviation is calculated by averaging the standard deviation on each trading day during the sample period. Minute-by-minute exchange rate data are sourced from histdata.com.

In trying to understand the true impact of terms of trade shocks, it is essential to break out the surprise component from the anticipated component of GDT price announcements. Exchange rates are forward looking, and any expected news about economic fundamentals should be incorporated into the price in advance. Thus, the relevant regression to identify β is:

$$\Delta e_a = \alpha + \beta s_a + \varepsilon_a, \quad (3)$$

where s denotes the surprise component of the announcement. Quantifying the response requires the ability to calculate the expected WMP price and how it differs from the actual WMP price.

Fortunately, this setting provides a simple way to calculate market expectations of the WMP auction price: WMP futures, as each month's WMP futures contract settles to the average price from the two GDT auctions that month.¹⁶ If I assume futures market participants are risk-neutral and have rational expectations, then the expected auction prices at auction $n \in \{1, 2\}$ of month m , which is on day t , is given by:

$$\mathbb{E}_{t-1}(P_{WMP,t}^{n=1,m}) = F_{WMP,t-1}^m$$

$$\mathbb{E}_{t-1}(P_{WMP,t}^{n=2,m}) = 2 \times F_{WMP,t-1}^m - P_{WMP,t-1}^{n=1,m}$$

where P_{WMP} denotes the WMP auction price, and $F_{WMP,t-1}$ denotes the last WMP futures price of the day before the auction.¹⁷ I assume that ahead of the first auction, participants expect the same price for the two auctions that month. Ahead of the second auction, the other relevant price has already been determined, so the calculation does not require any assumptions.¹⁸ Daily WMP futures prices were provided by New Zealand's Exchange (NZX), and historical auction prices and quantities were provided by GDT.

Given the expected WMP price, I can back out the surprise component of the WMP auction price for an auction a .¹⁹

$$s_a = \ln(P_a^{WMP}) - \ln(\mathbb{E}_{t-1}(P_a^{WMP})), \quad (4)$$

where s measures the percentage deviation of the ultimate auction price from the expected auction price. Summary statistics from October 2010, when WMP futures were launched, through the end of 2019 are given in Table 2. A few notable features stand out. First, the mean of the surprise metric is very close to zero (though a 95 percent confidence interval around the mean does not include zero). Second, the surprises can be quite large, from 33 percent more than the expected price to 19 percent less than the expected price. Panel (a) in Fig. 2 shows a graph of the surprise metric over time, and Panel (b) shows its distribution.

These surprises come from buyer demand, as supply is essentially fixed a few days in advance of the auction. Fonterra is the primary seller at the GDT auctions, supplying 99 to 100 percent of the whole milk powder sold, and it publishes auction supply forecasts three trading days before the auction. As Figure A.2 shows, auction supply rarely deviates from the published forecasted amount.^{20,21} This fact is important in supporting the exclusion restriction, which holds here as long as auction surprises affect

¹⁶ Specifically, the futures settle to the average contract period 2 (CP2) price. In keeping with the contract period definition change in September 2011, the futures settled to the CP1 price up through the September 6, 2011, auction. The contract is cash-settled the day after the second auction.

¹⁷ Because GDT auctions happen at noon GMT, which is overnight New Zealand time, there is an eight- to nine-hour gap between the closing time of the WMP futures market and the start of the auction. The expectation calculation relies on the assumption that there is no new information after the futures market closing time. Figure A.3 clarifies this timeline.

¹⁸ As a robustness check in the appendix, I use only the subset of auctions that were second in the month.

¹⁹ There are four days when GDT did not publish WMP prices: July 15, 2015; August 4, 2015; November 3, 2015; and March 7, 2017. I exclude these days from my analysis, as I cannot calculate the surprise metric. Prices were not published because bidding during the auction was insufficient to cause the announced price to rise above the starting price. For the second auctions in August 2015, November 2015, and March 2017, I calculate the expected auction price as

$$\mathbb{E}_{t-1}(P_{WMP,t}^{n=2,m}) = F_{WMP,t-1}^m$$

²⁰ Specifically, on 94 of the 188 days for which I have the supply forecast, the supply was exactly equal to the forecasted supply, and the appendix shows how my baseline estimates do not change when I restrict my sample to days when supply was perfectly predicted.

²¹ It is possible that there could be supply shocks to non-GDT supply, which could impact buyer demand at the GDT auctions. For this not to be an issue, I need the added assumption that Fonterra sells a constant proportion of its total supply at GDT auctions. While the amount that Fonterra sells at GDT auctions

Table 2
Summary statistics for the surprise metric.

Observations	217
Mean	−0.010
95% CI of mean	[−0.018, −0.002]
Std. dev.	0.057
Min	−0.192
Max	0.330
Mean of absolute value	0.041

The surprise metric is calculated per Eq. (4) using auction price data from Global Dairy Trade and futures data from New Zealand's Exchange.

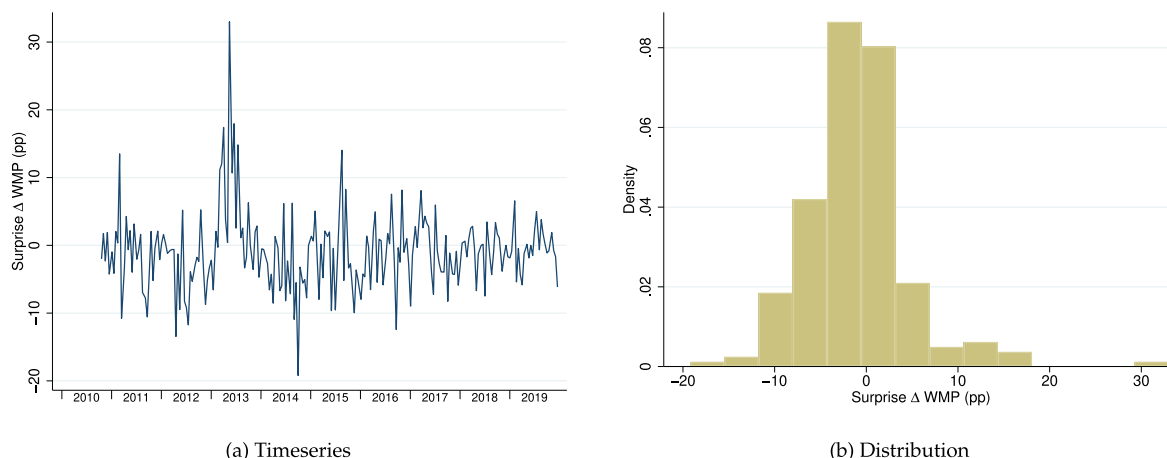


Fig. 2. Surprise metric timeseries and distribution. The surprise metric is calculated per Eq. (4) using auction price data from Global Dairy Trade (GDT) and futures data from New Zealand's Exchange (NZX).

the New Zealand dollar only through the WMP price. Surprises due to production shocks (such as weather events) might reveal information about the broader New Zealand economy, violating the exclusion restriction.

Furthermore, demand on the GDT auctions comes almost entirely (if not entirely) from foreign participants, with a significant portion coming from China. This is key support for the exclusion restriction. While GDT does not release data on the percentage of demand from outside New Zealand, less than 5 percent of the WMP produced by Fonterra is consumed domestically, so it should follow that a similarly small share of the product sold at GDT auctions is sold to domestic buyers.²² While the amount demanded by each country is not public, north Asia, which includes China, accounts for the majority of demand (Figure A.1). It receives 55 percent of all WMP auction supply on average over the course of my sample, though this amount is highly variable, ranging from 12 to 91 percent with an overall standard deviation of 15.7 percent (Table A.1).²³

These foreign auction buyers, however, are often excluded from the futures markets. Capital controls in China, as well as other countries in Southeast Asia, prevent auction buyers in these countries from engaging in the futures market unless they have offshore offices. Additionally, NZX derivatives brokers serve only clients of a certain size, and none of the 50 official global Fonterra resellers that buy on the GDT platform is large enough to work with these brokers. Thus, demand shocks from these participants will only be revealed at auction, rather than informing futures prices, enabling auction price surprises.

Although demand comes primarily from foreign buyers, and although these buyers are more demand-informative than futures market participants, we might worry that demand could be reflecting expectations of future New Zealand supply, which is reflective of New Zealand economic conditions. This worry is mitigated by the fact that it is unlikely foreign buyers have private information about future New Zealand supply, especially since futures market participants include New Zealand dairy suppliers. Actors from across the entire WMP supply chain participate in the WMP futures market, including companies with contracts that link prices to the GDT auctions but that have not been approved to participate in GDT auctions. This includes certain non-Fonterra dairy blenders and processors in New Zealand that sell abroad. It seems unlikely that a foreign GDT buyer would know more about future New Zealand supply than these companies. While their auction demand may reflect expectations about future WMP supply from locations

ranges from 18 to 32 percent annually over the course of my study, this proportion is auto-correlated on a quarterly basis. The auto-correlation coefficient is 0.56, which is significant at the 0.001 percent level. Note that the auction price is based on the matching of auction demand to auction supply. Additionally, many of the off-auction sales are made within the 24-hour period at a prespecified spread to the GDT auction price.

²² Anecdotally, the percentage sold domestically at GDT auctions is quite smaller.

²³ In an interview with me, a former dairy analyst for NZX noted, "Demand is driven by a small number of Chinese buyers".

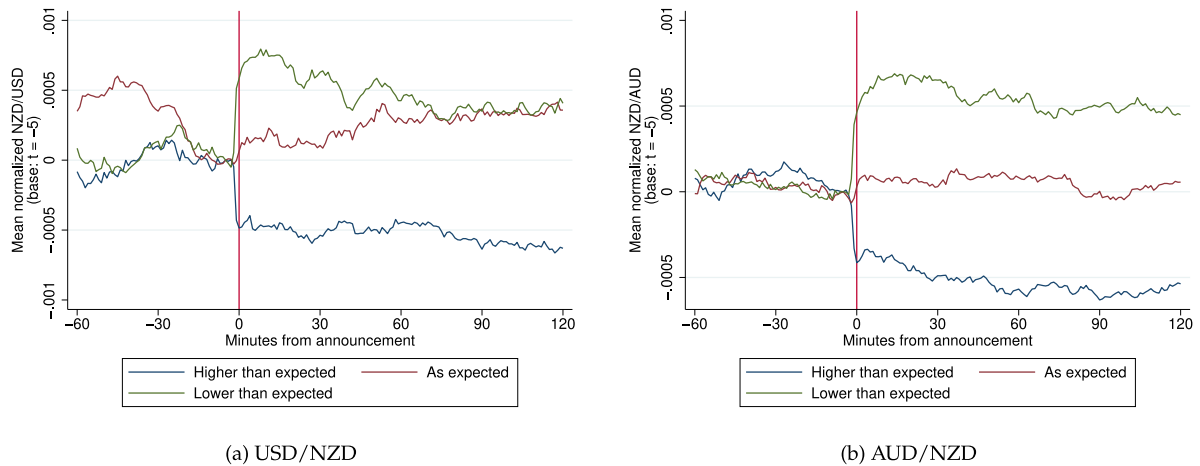


Fig. 3. Mean exchange rate changes by surprise group tercile. This figure shows that on auction days with negative whole milk powder (WMP) price surprises, the New Zealand dollar tends to depreciate immediately following the announcement, while on auction days with positive WMP price surprises, the New Zealand dollar tends to immediately appreciate following the announcement. Auction dates are split into terciles based on the surprise metric for the auction that day. I calculate the normalized exchange rate for each group by, first, taking the average exchange rate for the group at a given number of minutes around the announcement and, second, calculating the percent deviation from the mean exchange rate five minutes before the announcement. Minute-by-minute exchange rate data come from histdata.com. The surprise metric is calculated using auction price data from Global Dairy Trade (GDT) and futures data from New Zealand's Exchange (NZX). Auction announcement times were collected from Bloomberg.

other than New Zealand, this does not violate the exclusion restriction. As long as the impact of future domestic supply on demand is fully reflected in futures prices, it should not affect the surprise metric, and the exclusion restriction should hold.²⁴

Additionally, Appendix B provides evidence that WMP price surprises are milk-specific; they do not reveal information about Chinese growth, which would have other knock-on effects on non-WMP New Zealand exports.

5. Results of elasticity identification

To assess the effect of WMP price shocks on the New Zealand exchange rate, I start by splitting auction dates into terciles depending on the surprise metric and then examining whether the exchange rate responds differently on the three categories of days. Fig. 3 shows that on auction days with negative WMP price surprises, the New Zealand dollar tends to depreciate immediately following the announcement, whereas on auction days with positive WMP price surprises, the New Zealand dollar tends to appreciate immediately following the announcement. The announcement contains news that immediately moves the exchange rate in the direction we would anticipate.

I report my results for my baseline specification (Eq. (3)) in Table 3. Columns 1 and 3 show that the New Zealand dollar appreciates 0.010 percent against the US dollar and 0.009 against the Australian dollar for a 1 percent increase in the surprise component of WMP auction prices (a decrease in the exchange rate denotes an appreciation).²⁵ This effect is significant at the 1 percent level. Exchange rates *do* respond to terms of trade shocks. Specifically, when the export price index suddenly and unexpectedly increases, the exchange rate appreciates. Fig. 4 shows these results graphically in scatter plots to demonstrate that they are not driven by a single outlier event.

These movements are sizable compared with the mean daily standard deviation of the New Zealand dollar. Scaling by the mean USD/NZD exchange rate, a 0.010 percent movement corresponds to a 0.014 move in the exchange rate for a 1 percent change in the surprise component of WMP auction prices. Scaling by the mean absolute value of the surprise metric, on average we will see a roughly 0.055 move in the USD/NZD exchange rate after an auction announcement—quite a bit larger than the mean daily standard deviation of 0.0029.²⁶

²⁴ Fonterra also forecasts supply for an additional four auctions (over the ensuing two months) when it forecasts supply for the upcoming auction. Thus, all market participants know Fonterra's forecasts for the next two months of auctions a few days before each auction, and futures markets should take this information into account.

²⁵ Note that WMP auction prices are denominated in USD. In order to calculate the elasticity of exchange rates to WMP prices denominated in New Zealand dollars, note

$$\Delta WMP^{NZD} = \Delta WMP^{USD} - \Delta e.$$

For a 1 percent increase in USD WMP prices, the exchange rate appreciates 0.01 percent, and NZD WMP prices increase by 1.01 percent. Thus, the exchange rate appreciates $\frac{0.01}{1.01} = 0.0099\%$ for a 1 percent increase in NZD WMP prices.

²⁶ A similar statement applies to the AUD/NZD exchange rate. Scaling by the mean exchange rate, a 0.009 percent movement corresponds to a roughly 0.010 move in the exchange rate for a 1 percent change in the surprise component of WMP auction prices. Scaling by the mean absolute value of the surprise metric,

Table 3

The impact of surprise WMP price changes on the New Zealand dollar within 15 min after the announcement.

	Cumulative abnormal Δe (1) NZD/USD	(2) NZD/USD	(3) NZD/AUD	(4) NZD/AUD
Surprise Δ WMP	−0.010*** (0.002)	−0.012*** (0.002)	−0.009*** (0.001)	−0.011*** (0.002)
Expected Δ WMP		−0.004* (0.002)		−0.005*** (0.002)
Constant	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Observations	212	209	212	209
R2	0.090	0.104	0.167	0.183

Columns (1) and (3) report the results of the baseline specification in Eq. (3). Columns (2) and (4) control for the expected change in whole milk powder (WMP) auction prices. The cumulative abnormal exchange rate is calculated using an event window of 5 min before the announcement time through 15 min after the announcement time and an estimation window of 60 before the announcement time through 5 min before the announcement time. Coefficients are displayed with standard errors in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively. The surprise and expected WMP price changes are calculated using auction price data from Global Dairy Trade (GDT) and futures data from New Zealand's Exchange (NZX). Minute-by-minute exchange rate data are sourced from histdata.com. Auction announcement times were collected from Bloomberg.

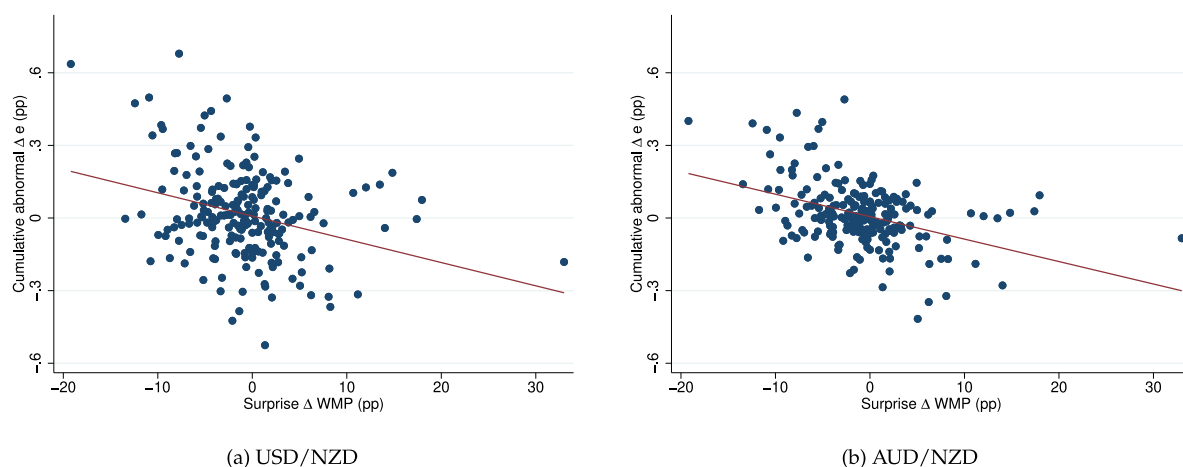


Fig. 4. Correlation between surprise WMP price changes and NZD exchange rate returns within 15 min after auction announcements. This figure shows the graphical results of the baseline specification in Eq. (3). Minute-by-minute exchange rate data come from histdata.com. The surprise metric is calculated using auction price data from Global Dairy Trade (GDT) and futures data from New Zealand's Exchange (NZX). Auction announcement times were collected from Bloomberg.

As discussed in Section 2, we should expect exchange rates to appreciate one-for-one with permanent terms of trade improvements. Because WMP makes up 6 to 11 percent of New Zealand's total exports over the sample period, if the surprise increase in WMP prices were permanent, we should expect exchange rates to appreciate 0.06 to 0.11 percent for a 1 percent WMP price surprise. It would seem markets do not expect these WMP surprises to be permanent. This is borne out in the data: WMP prices have an autocorrelation coefficient of 0.97 between auctions over the course of the sample.²⁷ This mean-reversion is a documented feature of agriculture commodity prices (Peterson et al., 1992; Bessembinder et al., 1995; Wang and Tomek, 2007).²⁸ That said, the resulting nominal exchange rate appreciation can be persistent even if the initial external demand shock is temporary. Appendix F demonstrates this result using impulse responses based on a model of this mechanism.

Columns 2 and 4 in Table 3 show that controlling for the expected change in WMP auction prices increases the magnitude of the unanticipated component slightly, to 0.012 percent and 0.011 percent for the USD and AUD exchange rates, respectively. I discuss expected changes more in Appendix C.

on average we will see a roughly 0.042 move in the AUD/NZD exchange rate after an auction announcement. Compare this with the mean daily standard deviation of 0.0015.

²⁷ This does not rule out the possibility of a persistent component to WMP prices. Both cyclical and structural elements drove WMP price movements during my sample period. The cyclical factors involved supply: a 2013 drought and the strong supply response. The structural factors involved Chinese growth as well as a changing milk market in China as part of the government's efforts to improve milk quality. Because my event study setting controls for supply, price shocks reflect shocks to demand, which are more likely to be indicative of structural changes to the WMP market.

²⁸ In standard commodity models, prices of storable commodities follow a random walk (Deaton and Laroque, 1992). That said, WMP is not infinitely storable, and recommended storage is 24 months. The documented mean-reversion in agriculture markets may be explained by a supply response to price movements.

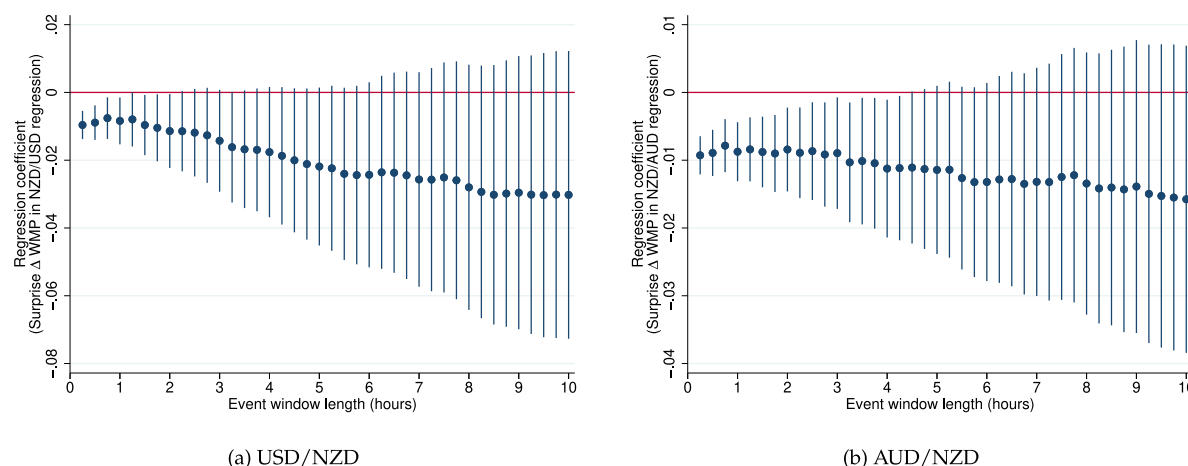


Fig. 5. The effect of surprise increases in WMP prices on the NZD exchange rate with a lengthening event window. This figure shows the results of the baseline specification in Eq. (3) with event windows that gradually lengthen by 15 min. Bars denote 95 percent confidence intervals. It is apparent that standard errors become increasingly larger as the event window lengthens. Minute-by-minute exchange rate data come from histdata.com. The surprise metric is calculated using auction price data from Global Dairy Trade (GDT) and futures data from New Zealand's Exchange (NZX). Auction announcement times were collected from Bloomberg.

To understand the benefits of my tight event window and provide evidence for the usefulness of a high-frequency event study in this context, I run the specifications in Columns (1) and (3) from Table 3 with event windows that gradually lengthen by 15 min. Fig. 5 shows how standard errors become increasingly larger after the shortest event window such that the response no longer seems significant. This points to the benefit of high-frequency identification with a short event window for studying exchange rate responses: Exchange rates are subject to a lot of noise.

However, we can see that the point estimate drifts such that the magnitude increases with a longer event window. Although it is important not to over-interpret this result due to the rising standard errors, this is a pattern similar to that seen in the post-earnings-announcement drift literature.^{29,30} Markets may take time to process the information from the auction, and the response within the first 15 min after the event may not be the entire response. This is especially likely when we consider that auctions take place overnight from a New Zealand perspective. This points to an efficiency-bias trade-off: A wider event window may give us a better sense of the actual magnitude of the exchange rate response, while a tighter window guarantees less noise and smaller standard errors.

Although exchange rates are quite noisy, which points to the benefits of high-frequency event studies to identify and detect exchange rate responses, GDT auction days are correlated with higher realized volatility of the New Zealand dollar. Specifically, the standard deviation of the minute-by-minute AUD/NZD exchange rate is 0.00054 higher on auction days than non-auction days, which corresponds to 36 percent of the mean daily standard deviation (0.0015). For the USD/NZD exchange rate, it is 0.00090 higher, which corresponds to 31 percent of the mean standard deviation (0.0029).

6. Conclusion

This paper provides causal evidence that terms of trade shocks *do* impact exchange rates. Specifically, in keeping with standard theory, export price increases cause exchange rate appreciations (and vice versa). The New Zealand setting serves as a well-identified natural experiment, but the result is important to all countries whose exports rely on a small number of commodities. As policymakers in these countries consider exchange rate policy, it is important for them to know the degree to which commodity prices will impact their exchange rate.

My focus here is on the impact of terms of trade shocks; however, WMP price surprises may additionally move the exchange rate if they prompt an imperfectly intermediated capital flow. Because Fonterra fully converts USD sales revenues into NZD, unexpected WMP price changes necessarily respond in a capital flow. That said, given how small this flow would be compared to New Zealand GDP, it is unlikely to explain the observed exchange rate response. Overall, results support the exchange rate moving due to surprise price movements in the export commodity, which market participants do not expect to be permanent.

More generally, this paper demonstrates how well-identified natural experiments can help advance our understanding of exchange rate determination. Seeking out additional natural experiments may be a fruitful avenue for future research.

²⁹ See Fink (2021) for a review.

³⁰ I further consider the possibility of pre-announcement drift in Appendix B. Specifically, Figure A.4 does not support pre-announcement drift.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hillary received financial support from the Weatherhead Center for International Affairs and the Institute for Quantitative Social Science at Harvard while working on this research. She also received financial support in the forms of grants and stipends from the Harvard Kennedy School as a Ph.D. student working on this research. She currently works for the Federal Reserve Bank of Boston. The views expressed here are solely those of the author and do not reflect the opinions of the Federal Reserve Bank of Boston, or the Federal Reserve System.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jinteco.2025.104080>.

Data availability

Got Milk? The Effect of Export Price Shocks on Exchange Rates (Original data) (Mendeley Data)

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